



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 - 2020)

NEET : PRACTICE PAPER

TARGET : PRE-MEDICAL - 2020

Test Type : PT # 06

FULL SYLLABUS

Test Pattern : NEET

TEST DATE : 15-04-2020

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is **3 Hours** and Questions Paper Contains **180** Questions. The Max. Marks are **720**.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while **1 mark will be deducted for every wrong answer**. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of **Blue / Black Ball Point Pen only** as the correct answer(s) of the question attempted.
8. **Use of Pencil is strictly prohibited.**

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with **Paper code** and Your **Form No.**

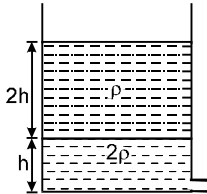
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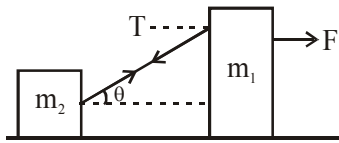
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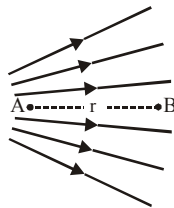
BEWARE OF NEGATIVE MARKING

- $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vector such that $\vec{a} + \vec{b} - \vec{c} = 0$, then the angle between $\vec{a}$ and $\vec{b}$ is :-
 (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{2}$ (4) $\frac{2\pi}{3}$
- A far sighted person has a near point of 60 cm. What power lens should be used for eye glasses such that the person can read this book at a distance of 25 cm. :-
 (1) -6.66 D (2) +6.66 D
 (3) +2.33 D (4) -2.33 D
- Modulation is the process of superposing:-
 (1) Low frequency audio signal on high frequency waves
 (2) Low frequency radio signal on low frequency audio waves
 (3) High frequency audio signal on low frequency radio waves
 (4) Low frequency audio signal on low frequency radio waves
- A parallel plate condenser with a dielectric of dielectric constant K between the plates has a capacity C and is charged to a potential V volts. The dielectric slab is slowly removed from between the plates and then reinserted. The net work done by the system in this process is:-
 (1) $\frac{1}{2}(K-1)CV^2$ (2) $CV^2 (K-1)/K$
 (3) $(K-1)CV^2$ (4) Zero
- If astronomical unit, parsec and light year deacted by A, B and C respectively then correct order of their magnitude is :-
 (1) $A > B > C$ (2) $A < B < C$
 (3) $A < C < B$ (4) $A > C > B$
- A photon of 1.6×10^{-13} Joules is absorbed by a material under special circumstances. The correct statement is:-
 (1) Electrons of the atom of absorbed material will go the higher energy states
 (2) Electron and positron pair will be created
 (3) Only positron will be produced
 (4) Photoelectric effect may occur and electron may be produced
- Radius of the sun is 7×10^8 m and the energy radiated by it is 38×10^{28} W. Calculate the magnitude of energy propagation poynting vector (i.e., energy flowing per second per unit area) on the surface of sun:-
 (1) 6.17×10^8 W/m² (2) 6.17×10^9 W/m²
 (3) 6.17×10^{10} W/m² (4) 6.17×10^{11} W/m²
- The kinetic energy needed to project a body of mass m from the earth's surface (radius R) to infinity is-
 (1) $\frac{mgR}{2}$ (2) $2mgR$ (3) mgR (4) $\frac{mgR}{4}$
- A particle is dropped from rest and another particle is thrown downward simultaneously with initial speed u, then :-
 (1) Time after which their separation becomes h is $\frac{h}{u}$
 (2) Their relative velocity is u
 (3) Their relative acceleration is always zero
 (4) All of these
- The X-ray wavelength of L_{α} line of platinum ($Z = 78$) is $1.30 \text{ \AA}$. The X-ray wavelength to L_{α} line of Molybdenum ($Z = 42$) is ($\phi = 7.4$):-
 (1) $5.41 \text{ \AA}$ (2) $4.20 \text{ \AA}$
 (3) $2.70 \text{ \AA}$ (4) $1.35 \text{ \AA}$
- Amount of work done in moving a charge q_0 from centre of ring to an axial point at $x = \sqrt{8}a$ will be if charge given to ring is Q and 'a' being the radius:-
 (1) $\frac{Qq_0}{6\pi\epsilon_0 a}$ (2) $-\frac{Qq_0}{6\pi\epsilon_0 a}$ (3) Zero (4) $\frac{Qq_0}{4\pi\epsilon_0 a}$
- The velocity of the liquid coming out of a small hole of a large vessel containing two different liquids of densities 2ρ and ρ as shown in figure is

 (1) $\sqrt{6gh}$ (2) $2\sqrt{gh}$ (3) $2\sqrt{2gh}$ (4) $\sqrt{gh}$

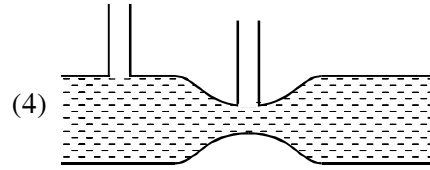
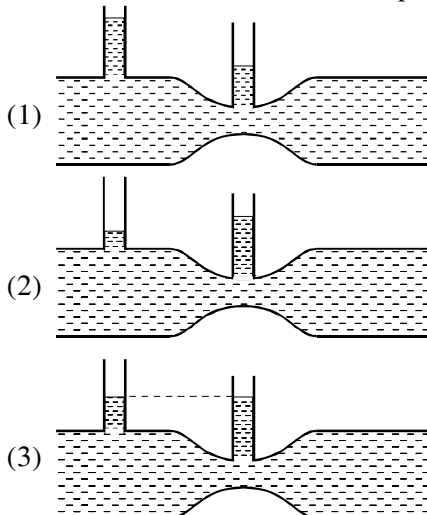
13. In given system, tension in string connecting two bodies will be [All surfaces are frictionless] :-



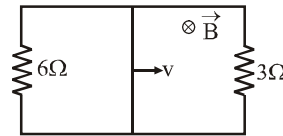
- (1) $\frac{m_2 F}{m_1 + m_2}$ (2) $\frac{m_2 F \cos \theta}{m_1 + m_2}$
(3) $\frac{m_2 F}{(m_1 + m_2) \cos \theta}$ (4) $\frac{m_2 F}{(m_1 + m_2) \tan \theta}$
14. The sun radiates energy in all directions. The average radiations received on the earth surface from the sun is 1.4 kilowatt/m^2 . The average earth-sun distance is 1.5×10^{11} metres. The mass lost by the sun per day is (1 day = 86400 seconds):-
- (1) $4.4 \times 10^9 \text{ kg}$ (2) $7.6 \times 10^{14} \text{ kg}$
(3) $3.8 \times 10^{12} \text{ kg}$ (4) $3.8 \times 10^{14} \text{ kg}$
15. Figure shows the electric lines of force emerging from a charged body. If the electric field at A and B are E_A and E_B respectively and if the displacement between A and B is r then :-



- (1) $E_A > E_B$ (2) $E_A < E_B$
(3) $E_A = \frac{E_B}{r}$ (4) $E_A = \frac{E_B}{r^2}$
16. For a fluid which is flowing steadily, the level in the vertical tubes is best represented by

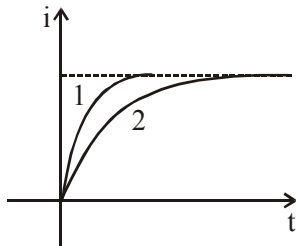


- (4)
17. A boy swinging on a swing from rest. The power delivered to the boy when he is moving with velocity v away from mean position and string makes angle θ with vertical:-
(1) $mg v \cos \theta$ (2) $-mg v \cos \theta$
(3) $mg v \sin \theta$ (4) $-mg v \sin \theta$
18. A rectangular loop with a sliding connector of length $\ell = 1.0 \text{ m}$ is situated in a uniform magnetic field $B = 2 \text{ T}$ perpendicular to the plane of loop. Resistance of connector is $(r) 2 \Omega$. Two resistors of 6Ω and 3Ω are connected as shown in figure. The external force required to keep the connector moving with a constant velocity $v = 2 \text{ m/s}$ is

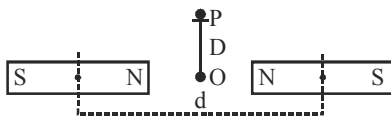


- (1) 6 N (2) 4 N (3) 2 N (4) 1 N
19. A positively charged ($+q$) particle of mass m has kinetic energy K enters vertically downward in a horizontal field of magnetic induction $\vec{B}$. The acceleration of the particle is : (Neglect gravity)
- (1) $qB \sqrt{\frac{2K}{m}}$ (2) $\frac{qB \sqrt{2K}}{(m)^{3/2}}$
(3) $\frac{2qB}{(m)^{3/2}} \sqrt{2K}$ (4) $2qB \sqrt{\frac{2K}{m}}$
20. One mole of an ideal gas at a temperature $T_1 \text{ K}$ expands slowly according to the law $\frac{p}{V}$ is constant. Its final temperature is $T_2 \text{ K}$. The work done by the gas is
(1) $R(T_2 - T_1)$ (2) $2R(T_2 - T_1)$
(3) $\frac{R}{2}(T_2 - T_1)$ (4) $\frac{2R}{3}(T_2 - T_1)$
21. Choose the correct statement :-
(1) The net acceleration of a particle in circular motion is always towards the centre of the circle
(2) The acceleration vector of a particle in uniform circular motion averaged over one cycle is zero
(3) It is not possible to revolve a particle in circular path with constant acceleration
(4) Both (2) & (3)

22. When a certain circuit consisting of a constant e.m.f. E , an inductance L and a resistance R is closed, the current in it increases with time according to curve 1. After one parameter (E , L or R) is changed, the increase in current follows curve 2 when the circuit is closed second time. Which parameter was changed and in what direction



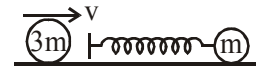
- (1) L is increased (2) L is decreased
(3) R is increased (4) All of these
23. Two identical bar magnets are fixed with their centres at a distance d apart. A stationary charge Q is placed at P in between the gap of the two magnets at a distance D from the centre O as shown in the figure



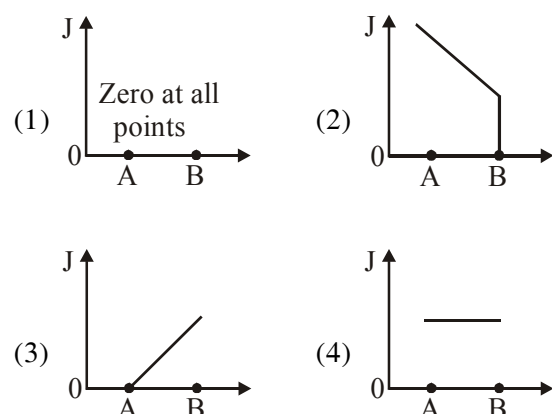
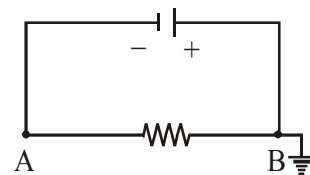
The force on the charge Q is :-

- (1) Zero
(2) Directed along OP
(3) Directed along PO
(4) Directed perpendicular to the plane of paper
24. Four cubes of ice at -10°C each one gm is taken out from the refrigerator and are put in 150 gm of water at 20°C . The temperature of water when thermal equilibrium is attained. Assume that no heat is lost to the outside and water equivalent of container is 46 gm. (Specific heat capacity of water = $1 \text{ cal/gm}^\circ\text{C}$, Specific heat capacity of ice = $0.5 \text{ cal/gm}^\circ\text{C}$, Latent heat of fusion of ice = $80 \text{ cal/gm}^\circ\text{C}$)
- (1) 0°C (2) -10°C
(3) 17.9°C (4) None

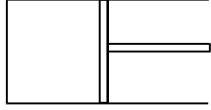
25. A particle of mass $3m$ is moving with speed v collides with another particle of mass m as shown in figure, maximum compression in spring is $[K = \text{force constant of spring}] :-$



- (1) $\sqrt{\frac{3mv^2}{4K}}$ (2) $\sqrt{\frac{4mv^2}{3K}}$
(3) $\sqrt{\frac{3mv^2}{16K}}$ (4) $\sqrt{\frac{7mv^2}{8K}}$
26. $\frac{2.5}{\pi} \mu\text{F}$ capacitor and 3000-ohm resistance are joined in series to an ac source of 200 volt and 50 sec^{-1} frequency. The power factor of the circuit and the power dissipated in it will respectively be :-
- (1) 0.6, 0.06 W (2) 0.06, 0.6 W
(3) 0.6, 4.8 W (4) 4.8, 0.6 W
27. A battery is connected to a uniform resistance wire AB and B is earthed. Which one of the graphs below shows how the current density J varies along AB :-



28. n moles of a gas filled in a container at temperature T is in thermodynamic equilibrium initially. If the gas is compressed slowly and isothermally to half its initial volume the work done by the atmosphere on the piston is :

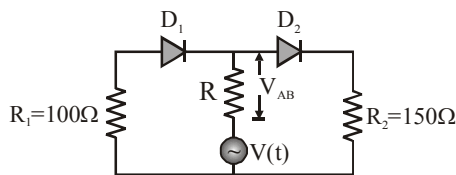


- (1) $\frac{nRT}{2}$ (2) $-\frac{nRT}{2}$
(3) $nRT \left(\ln 2 - \frac{1}{2} \right)$ (4) $-nRT \ln 2$

29. If a sphere is rolling down a rough incline with sliding :-

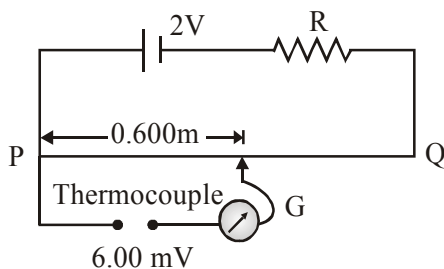
- (1) After some time it will start pure rolling
(2) After some time it may start pure rolling
(3) It can never start pure rolling
(4) Both (2) & (3)

30. In the circuit given below, $V(t)$ is the sinusoidal voltage source, voltage drop $V_{AB}(t)$ across the resistance R is :-



- (1) Is half wave rectified
(2) Is full wave rectified
(3) Has the same peak value in the positive and negative half cycles
(4) Has different peak values during positive and negative half cycle

31. Figure shows a simple potentiometer circuit for measuring a small e.m.f. produced by a thermocouple. The meter wire PQ has a resistance 5Ω and the driver cell has an e.m.f. of $2V$. If a balance point is obtained 0.600 m along PQ when measuring an e.m.f. of 6.00 mV, what is the value of resistance R :-



- (1) 995Ω (2) 1995Ω
(3) 2995Ω (4) None of these

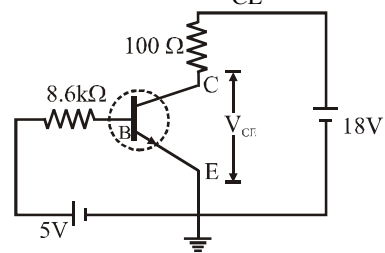
32. The acceleration of a particle moving along x -axis is $a = -100x + 50$. It is released from $x = 2$. Here 'a' and 'x' are in S.I units. The motion of particle will be :

- (1) periodic, oscillatory but not SHM.
(2) periodic but not oscillatory.
(3) oscillatory but not periodic.
(4) simple harmonic.

33. A parallel beam of white light falls on convex lens. Images of violet, yellow and red light are formed on the other side of the lens at a distance of 20 cm, 20.5 cm and 21.4 cm respectively. The dispersive power of the material of the lens will be :-

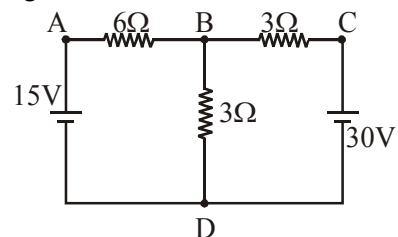
- (1) $14/205$ (2) $9/200$
(3) $5/214$ (4) $619/1000$

34. For the transistor circuit shown below, if $\beta = 100$, voltage drop between emitter and base is 0.7 V then value of V_{CE} will be :-



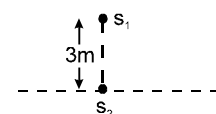
- (1) $10V$ (2) $5V$ (3) $13V$ (4) $0V$

35. In the circuit shown in figure, find the current through the branch BD:-



- (1) $5A$ (2) $0A$ (3) $3A$ (4) $4A$

36. S_1 & S_2 are two coherent sources of sound having no initial phase difference. The velocity of sound is 330 m/s. No minima will be formed on the line passing through S_2 and perpendicular to the line joining S_1 and S_2 , if the frequency of both the sources is :

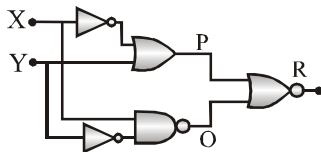


- (1) 50 Hz (2) 60 Hz
(3) 70 Hz (4) 80 Hz

37. In YDSE the Y co-ordinates of central maxima and 10th maxima are 2cm and 5cm respectively. When YDSE apparatus is immersed in a liquid of refractive index 1.5. The corresponding Y co-ordinates will be :-

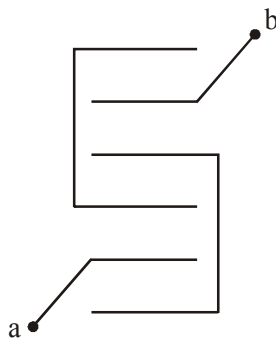
- (1) 4 cm, 7.5 cm (2) 3 cm, 6 cm
(3) 2 cm, 4 cm (4) 4/3 cm, 10/3 cm

38. Figure gives a system of logic gates. From the study of truth table it can be found that to produce a high output (1) at R, we must have :-



- (1) $X = 0, Y = 1$ (2) $X = 1, Y = 1$
(3) $X = 1, Y = 0$ (4) $X = 0, Y = 0$

39. Plates of area A are arranged as shown. The distance between each plate is d, the net capacitance is:-

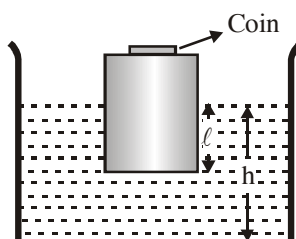


- (1) $\frac{\epsilon_0 A}{d}$ (2) $\frac{7\epsilon_0 A}{d}$ (3) $\frac{6\epsilon_0 A}{d}$ (4) $\frac{5\epsilon_0 A}{d}$

40. If the fundamental frequency of a pipe closed at one end is 512Hz. The fundamental frequency of a pipe of the same dimensions but open at both ends will be :

- (1) 1024 Hz (2) 512Hz
(3) 256 Hz (4) 128Hz

41. A wooden block with a coin placed on its top floats in water as shown in figure. The ℓ and h are shown there. After some time the coin falls into the water then :-

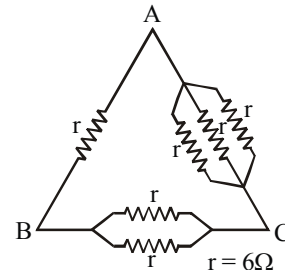


- (1) ℓ decrease and h increase
(2) ℓ increase and h decrease
(3) Both ℓ & h increase
(4) Both ℓ & h decrease

42. A ball moving with a velocity v strikes a wall moving towards the ball with a velocity u. An elastic impact lasts for t seconds then the mean elastic force acting on the ball is (Mass of the ball is M) :-

- (1) $\frac{2Mv}{t}$ (2) $\frac{M(v + 2u)}{t}$
(3) $\frac{2M(v + u)}{t}$ (4) $\frac{M(2v + u)}{t}$

43. Six resistances each of value $r = 6 \Omega$ are connected between points A, B and C as shown in the figure. If R_1 , R_2 and R_3 are the net resistance between A and B, between B and C and between A and C respectively, then $R_1 : R_2 : R_3$ will be equal to :-



- (1) 6 : 3 : 2 (2) 1 : 2 : 3
(3) 5 : 4 : 3 (4) 4 : 3 : 2

44. Four spheres A, B, C and D are of same radius but made of different metals. Their densities are in ratio 6 : 3 : 4 : 5 and specific heats are in ratio 2 : 5 : 4 : 6. These are initially kept at the same temperature and placed in the same surroundings. The sphere which has the slowest rate of cooling is :-

- (1) C (2) B (3) D (4) A

45. A square current carrying loop is placed in uniform magnetic field. If the magnetic force on one arm of the loop is $\vec{F}$ then net magnetic force on the remaining three arms of the loop:-

- (1) $3\vec{F}$ (2) $-3\vec{F}$ (3) $\vec{F}$ (4) $-\vec{F}$

46. 0.2 mole of HCl and 0.2 mole of barium chloride were dissolved in water to produce a 500 mL solution. The molarity of the Cl^- ions is :-

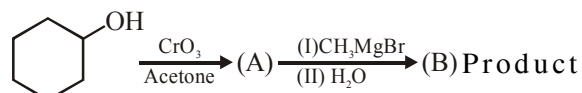
- (1) 0.06 M (2) 0.09 M
(3) 1.2 M (4) 0.80 M

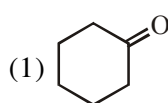
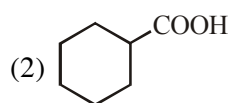
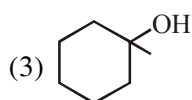
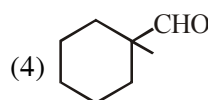
47. The relationship between osmotic pressure at 273K when 10 g glucose (p_1), 10g urea (p_2) and 10 g sucrose (p_3) are dissolved in 250 mL of water is:-

- (1) $P_1 > P_2 > P_3$
(2) $P_3 > P_1 > P_2$
(3) $P_2 > P_1 > P_3$
(4) $P_2 > P_3 > P_1$

48. Most stable allotropic form of phosphorous is :-

- (1) Red (2) White
(3) Black (4) Yellow

49.  (B) is:-

- (1)  (2) 
(3)  (4) 

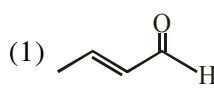
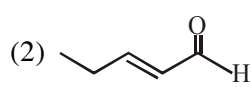
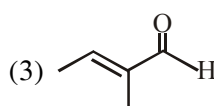
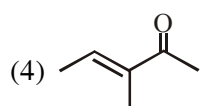
50. Find the incorrect match :-

- (1) Al_2Cl_6 : 3C-4e⁻ bond is present
(2) $\text{Al}_2(\text{CH}_3)_6$: All carbon atoms are sp^3 hybridised
(3) I_2Cl_6 : Non-planar
(4) Al_2Br_6 : Non-planar

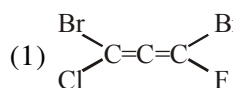
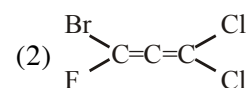
51. Crystal field stabilisation energy for high spin d^4 octahedral complex is :-

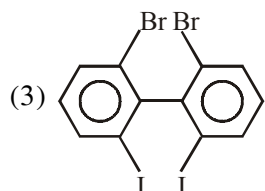
- (1) $-1.8 \Delta_0$ (2) $-1.6 \Delta_0 + P$
(3) $-1.2 \Delta_0$ (4) $-0.6 \Delta_0$

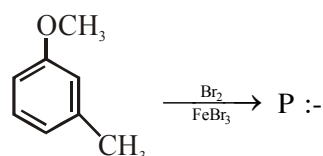
52. $\text{CH}_3\text{-CHO} + \text{CH}_3\text{-CH}_2\text{-CHO} \xrightarrow[\Delta]{\text{dil NaOH}} (\text{P})$
which product will not form :-

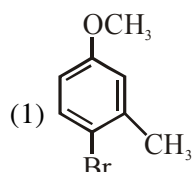
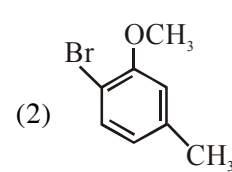
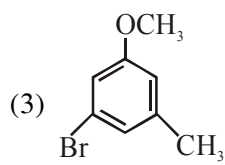
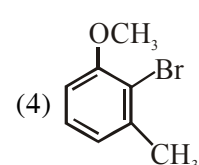
- (1)  (2) 
(3)  (4) 

53. Which is optically active:-

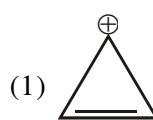
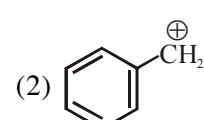
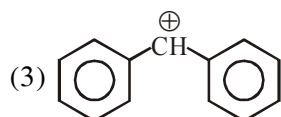
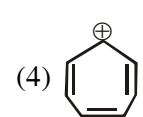
- (1)  (2) 

- (3)  (4) 1 and 3

54.  P :-

- (1)  (2) 
(3)  (4) 

55. Which carbocation is more stable in following:-

- (1)  (2) 
(3)  (4) 

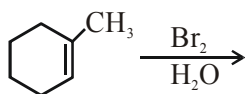
56. The body centered cubic cell of chromium has a length of 0.288 nm. Calculate the density of chromium (g/cm^3) : (Atomic weight : Cr = 52.0):-

- (1) 6.80 (2) 7.60
(3) 6.60 (4) 7.23

57. $\text{Na}_2\text{S}_2\text{O}_3$ is oxidised by I_2 to :-

- (1) Na_2S (2) Na_2SO_4
(3) NaHSO_3 (4) $\text{Na}_2\text{S}_4\text{O}_6$

58. The resistance of 1 M – CH₃COOH solution is 250 Ω, when measured in a cell of cell constant 125 m⁻¹. The molar conductivity, in Ω⁻¹ m² mol⁻¹ is
 (1) 5.0 × 10⁻⁴
 (2) 500
 (3) 2 × 10⁻³
 (4) 200
59. 25 g ethylene glycol is present in 100 g of water. The solution is cooled to -10°C. K_f for H₂O is 1.86 K kg mol⁻¹. The amount of ice separated on cooling is :-
 (1) 25 g (2) 50 g
 (3) 75 g (4) 20 g
60. Incorrect match is :-
 (1) [Co(en)₃]³⁺ ⇒ d²sp³, Diamagnetic, Optical isomerism
 (2) [Co(en)₂Cl₂]Br ⇒ Ionisation, Geometrical, Optical isomerism
 (3) [Fe(CN)₆]⁴⁻ ⇒ μ_m = 1.73 BM, d²sp³, Low spin
 (4) [Cr(NH₃)₆]³⁺ ⇒ μ_m = 3.82 BM, d²sp³, Inner orbital complex
61. A ionic solid A⁺B⁻ crystallises as a bcc structure. The distance between cation and anion in the lattice is 338 pm. The edge length of cell is:-
 (1) 338 pm (2) 390.3 pm
 (3) 292.7 pm (4) 507 pm
62. Which of the following is a major product of the reaction show below ?



- (1)
- (2)
- (3)
- (4)

63. $\text{N}_2\text{O}_5 \longrightarrow 2\text{NO}_2 + \frac{1}{2}\text{O}_2$;
 In a definite time interval the rate of decomposition of N₂O₅ is 1.8 × 10⁻³ mol L⁻¹ min⁻¹, then the rate of formation of O₂ will be
 (1) 1.8 × 10⁻³ mol L⁻¹ min⁻¹
 (2) 9 × 10⁻⁴ mol L⁻¹ min⁻¹
 (3) 3.6 × 10⁻³ mol L⁻¹ min⁻¹
 (4) 3.6 × 10⁻⁴ mol L⁻¹ min⁻¹
64. The wavelength associated with a golf ball weighing 200 g and moving at a speed of 5 m h⁻¹ is of the order:-
 (1) 10⁻¹⁰m (2) 10⁻²⁰m
 (3) 10⁻³⁰m (4) 10⁻⁴⁰m

65. 'σ' bond formed in :-
 (1) overlapping of d_{z²} + d_{z²} in z-axis
 (2) overlapping of d_{xy} + d_{xy} in y-axis
 (3) overlapping of d_{x²-y²} + d_{x²-y²} in z-axis
 (4) overlapping of d_{xy} + d_{xy} in z-axis

66.
 $\xrightarrow{\text{NaOI}}$ (B) + (C) Product (A) and (C) is :-

- (1)
- (2)
- (3)
- (4)

- 67.
- (I) (II) (III)

Which of the following order is correct for heat of Hydrogenation of these compounds :-

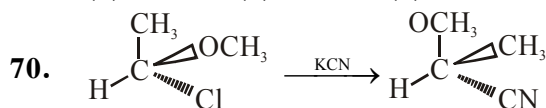
- (1) I > II > III (2) III > II > I
 (3) II > III > I (4) III > I > II

68. Which of the following is a biodegradable polymer :-

- (1) PMMA (2) Glyptal
(3) Nylon-2, 6 (4) PVA

69. What will be the molality of 2M H_2SO_4 solution having density 1.8 g/ml :-

- (1) 3 m (2) 1.25 m (3) 2.4 m (4) 4.8 m



Correct statement of the following reaction is :-

- (1) SN^2 , inversion configuration takes place
(2) SN^2 retention configuration takes place
(3) SN^2 retention product obtain
(4) SN^i , inversion configuration takes place

71. The rate constant for the first order reaction is given by :

$$\ln K = 14.34 - \frac{1.25 \times 10^4}{T}$$

Calculate the activation energy :

- (1) $2.393 \times 10^5 \text{ J}$ (2) $8.69 \times 10^4 \text{ J}$
(3) $1.18 \times 10^5 \text{ J}$ (4) $8.69 \times 10^5 \text{ J}$

72. What will be the osmotic pressure of 0.2 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution at 27°C ?

- (1) 24.6 atm (2) 49.2 atm
(3) 4.92 atm (4) 2 atm

73. Which statement is incorrect ?

- (1) In safety matches KClO_3 is used.
(2) Purest form of carbon allotrope is fullerene.
(3) Red phosphorous glows in dark
(4) $\text{H}_2\text{S} + \text{HNO}_3 \longrightarrow \text{Colloidal sulphur}$

74. What is the ratio of time periods (T_1/T_2) in second orbit of hydrogen atom to third orbit of He^+ ion ?

- (1) 8/27 (2) 32/27
(3) 27/32 (4) 8/9

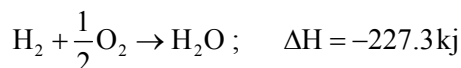
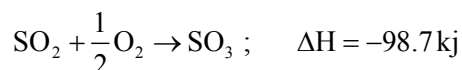
75. In the hydrolysis process of $\text{H}_2\text{S}_2\text{O}_8$ which is formed ?

- (1) H_2SO_5
(2) H_2O_2
(3) H_2SO_4
(4) All given compound formed

76. If the ratio of the rates of diffusion of the two gases A & B is 4 : 1, the ratio of their densities is :-

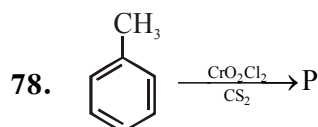
- (1) 1 : 16 (2) 1 : 4
(3) 1 : 8 (4) 1 : 2

77. If $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$; $\Delta H = -298.2 \text{ kJ}$

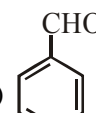
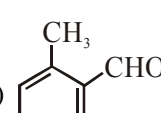
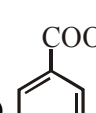
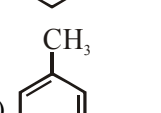


the heat of formation of H_2SO_4 will be :-

- (1) -754.4 kJ (2) +320.5 kJ
(3) -650.3 kJ (4) -433.7 kJ



Product (P) will be :-

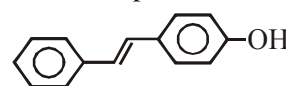
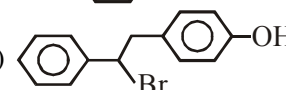
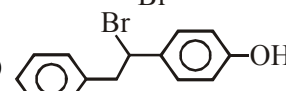
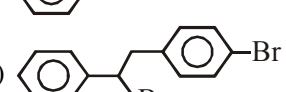
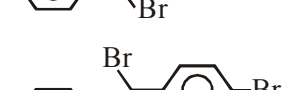
- (1)  (2) 
(3)  (4) 

79. Consider the following statements if they are true then use 'T' otherwise 'F' for false.

- (a) $\text{K}_2\text{CO}_3 > \text{BeCO}_3$ (Thermal stability)
(b) $\text{SiO}_2 > \text{CO}_2$ (Order of hardness)
(c) LiF is ionic but insoluble in water
(d) On heating NH_4NO_3 a gas is evolved which is used in dentistry.

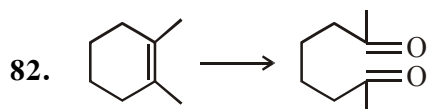
- (1) T T T T (2) T T F T
(3) T T F F (4) T F F T

80. The reaction of HBr with the following compound would produce :

- 
- (1) 
(2) 
(3) 
(4) 

81. The Ksp of PbCl_2 is 4×10^{-6} . Its solubility in 0.1 M NaCl solution is :-

- (1) 4×10^{-4} M (2) 4×10^{-5} M
(3) 4×10^5 M (4) 4×10^{-7} M



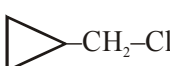
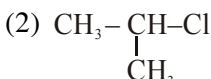
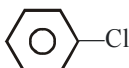
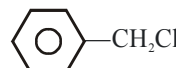
To carry out above conversion; which of the following reagent is used :

- (1) O_3/Zn (2) $\text{EtONa}/\text{EtOH}/\Delta$
(3) NaOCl (4) H^+

83. The SRP for Zn^{2+}/Zn , Ni^{2+}/Ni and Fe^{2+}/Fe are -0.76 V, -0.23 V and -0.44 V respectively. The reaction $\text{X} + \text{Y}^{2+} \longrightarrow \text{X}^{2+} + \text{Y}$ will spontaneous when :

- (1) $\text{X} = \text{Ni}$, $\text{Y} = \text{Fe}$
(2) $\text{X} = \text{Ni}$, $\text{Y} = \text{Zn}$
(3) $\text{X} = \text{Fe}$, $\text{Y} = \text{Zn}$
(4) $\text{X} = \text{Zn}$, $\text{Y} = \text{Ni}$

84. Which of the following is least reactive for $\text{S}_{\text{N}}2$ reaction ?

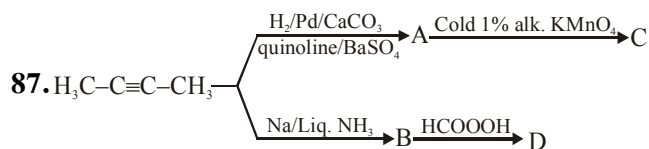
- (1)  (2) 
(3)  (4) 

85. If Z is compressibility factor than vander waal's equation of low pressure will be :-

- (1) $Z = 1 - \frac{\text{Pb}}{\text{RT}}$ (2) $Z = 1 + \frac{\text{Pb}}{\text{RT}}$
(3) $Z = 1 + \frac{\text{RT}}{\text{Pb}}$ (4) $Z = 1 - \frac{a}{\text{VRT}}$

86. Ferrocene is :-

- (1) $\text{Fe}(\eta^2\text{-C}_6\text{H}_5)_2$ (2) $\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2$
(3) $\text{Fe}(\eta^6\text{-C}_6\text{H}_6)_2$ (4) $\text{Fe}(\eta^3\text{-C}_3\text{H}_3)_2$



Relationship between (A,B) and (C,D) respectively are :-

- (1) Identical and Enantiomer
(2) Diastereomer and Identical
(3) Identical and Diastereomer
(4) Geometrical isomer and Diastereomer

88. When CuSO_4 is added to a solution of ammonia

- (1) Freezing point is lowered
(2) Freezing point is raised
(3) Boiling point is raised
(4) Both (1) and (3)

89. Correct statement is :-

- (1) $\text{F-F} > \text{Cl-Cl}$ bond dissociation energy
(2) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ thermal stability
(3) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ reducing power
(4) None of these

90. The π acid ligand which uses its 'd' orbital during synergic bonding in its complex compound is :-

- (1) NO^+ (2) CO
(3) H_2 (4) PR_3

91. Match the column :-

Column I		Column II		Column III	
(1)	Mutualism	(a)	Lianas	(i)	One partner is unaffected
(2)	Commensalism	(b)	Lichen	(ii)	non obligatory
(3)	Proto cooperation	(c)	Hermit crab-sea Anemone	(iii)	obligatory

- (1) Only 1 (2) 1 and 2
(3) Only 3 (4) All are incorrect

92. UN conference of parties (COP) on climate change in 2015 was held at ?

- (1) Doha (2) Durban (3) Lima (4) Paris

93. Read the following statements (A-D) carefully :-

[A] Genetic modification may improve tolerance to abiotic stress.

[B] Genetic modification increases nutritional value of food crops.

[C] Genetic modification helps to reduce in post-harvest losses.

[D] Genetic modification in animals provides facility to test vaccines before using on human.

How many of the above statements are correct ?

- (1) A, B and C (2) A and B
(3) A and D (4) All of the above

94. In glycolysis, pyruvic acid is formed :-

- (1) Only when O_2 is available
(2) Only during aerobic respiration
(3) In both aerobic and anaerobic respiration
(4) When only glucose is used as substrate

95. In meiosis, crossing over and terminalization of chiasmata respectively occur in ?

- (1) Pachytene & diplotene
(2) Pachytene & diakinesis
(3) Zygotene & pachytene
(4) Pachytene & zygotene

96. Class and stored food of brown algae is :-

- (1) Phaeophyceae & Floridean starch
(2) Phaeophyceae & Starch
(3) Phaeophyceae & Mannitol
(4) Rhodophyceae & Floridean starch

97. Which of the following have zygomorphic flower ?

- (1) Mustard, pea (2) Gulmohar, chilli
(3) Cassia, bean (4) Pea, datura

98. Which of the following groups of phylum show common body plan and symmetry ?

- (1) Annelida and ctenophora
(2) Coelenterata and flat worms
(3) Annelida and arthropoda
(4) Echinodermata and ctenophora

99. In the columns given below the differences between white and red muscles are mentioned. Find out the wrong one :-

	A (Red muscles)	B (White muscles)
(1)	Myoglobin content high	Myoglobin content low
(2)	Blood vessels are more extensive	Blood vessels are less extensive
(3)	Contraction more powerful	Contraction less powerful
(4)	Fatigue occur slowly	Fatigue occur quickly

100. How many of the following actions of sympathetic nerves are present on the heart ?

- (A) Increases heart beat.
(B) Increases cardiac output.
(C) Decrease the ventricular contraction.
(D) Decrease cardiac output.

- (1) One (2) Two (3) Three (4) Four

101. It is best for growing crop, it contain 50% sand and 50% silt, it hold maximum water. Which kind of this soil is ?

- (1) Sandy soil (2) Loamy soil
(3) Silt soil (4) Laterite soil

102. Which of the following is correct equation for Verhulst-Pearl logistic growth ?

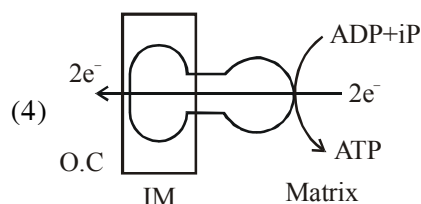
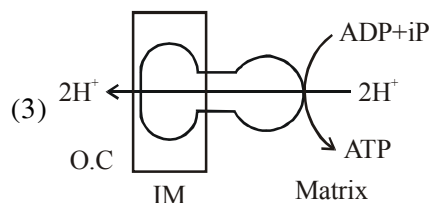
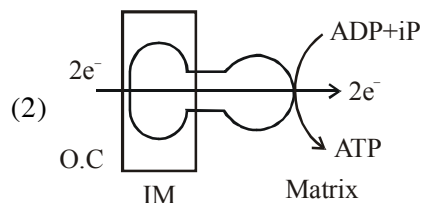
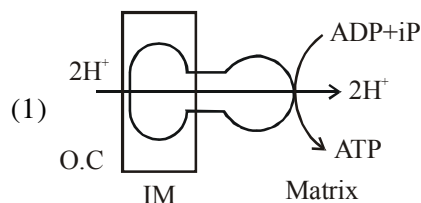
(1) $\frac{dN}{dt} = rN \left(\frac{N-K}{N} \right)$ (2) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$

(3) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$ (4) $\frac{dN}{dt} = rN \left(\frac{K}{K-N} \right)$

103. Large holes in 'Swiss cheese' are due to the production of large amount of :-

- (1) O_2 (2) CO_2
(3) Ethanol (4) (2) & (3) both

104. Which of the following diagrams represents ATP synthesis in mitochondria through chemiosmosis?



105. If in a cell the amount of DNA is $2c$ in metaphase then what will be the amount of DNA in anaphase, (x) in G_1 phase (y) and after M-phase (z) ?

- (1) (x) – $4c$, (y) – c , (z) – c
(2) (x) – $2c$, (y) – $2c$, (z) – $4c$
(3) (x) – $4c$, (y) – $4c$, (z) – $2c$
(4) (x) – $2c$, (y) – c , (z) – c

106. Which of the following statement is incorrect for DNA ?

- (1) DNA is a long polymer of deoxyribonucleoside monophosphate.
(2) A bacteriophage known as $\phi \times 174$ has 5386 base pair.
(3) Haploid content of human DNA is 3.3×10^9 bp.
(4) 5-methyl-uracil nitrogenous base present in DNA.

107. Which of the following match is **correct** ?

- (A) Endarch – Protoxylem toward the centre & metaxylem toward the periphery eq. stem & root
(B) Exarch – Metaxylem toward centre & protoxylem toward the periphery eq. root
(C) Xylem parenchyma – Consists of living cells and have lignified cell wall
(D) Tracheids – Cell are dead and without protoplasm

- (1) C, D (2) B, D (3) B, C, D (4) B, D

108. Which of the following animal show monocondylic skull :-

- (1) Frog (2) Birds
(3) Mammals (4) Rabbit

109. Consider the following four statements(a-d) and select the one which includes all **incorrect** ones:-

- (a) The essence of Darwinian theory of evolution is natural selection.
(b) Evolution is directed process in sense of determinism.

(c) The geological history of earth is not related with biological history of earth.

(d) During evolution the rate of appearance of new forms is linked with the life cycle.

- (1) a and b (2) b and c
(3) a and d (4) b and d

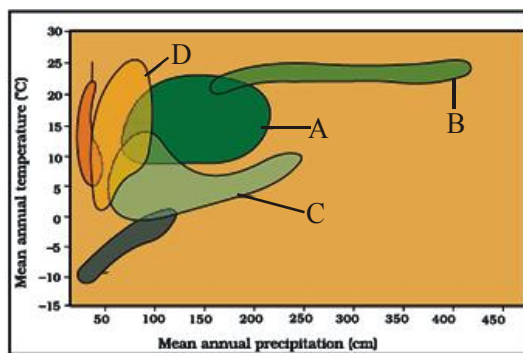
110. How many of the following is/are not correct about pneumonia :-

- (A) It is caused by *Salmonella typhi*.
(B) Symptoms of pneumonia include fever with chills, cough and headache.
(C) Infect alveoli of lungs.
(D) In severe cases intestinal perforation may occur.

Options :-

- (1) 3 (2) 2
(3) 1 (4) 4

111. On the basis of following digram most of the geographical region of india must be included in ?



- (1) C (2) A (3) B (4) D

112. Which of the following sentence is incorrect about multiple alleles ?

- (1) Multiple alleles can be found in population study
(2) Multiple alleles are formed by mutation
(3) In multiple allele system, a pollen grain has several alleles for a gene
(4) Multiple alleles are present at the same locus of a pair of homologous chromosomes

113. A colourblind man has a colourblind sister, but a normal brother then his mother and father would be:-

- (1) Both are colour blind
(2) Father normal but mother colourblind
(3) Father normal but mother carrier
(4) Father colour blind but mother carrier

114. Which relationship is considered correct for a solution at atmospheric pressure ?

- (1) $\psi_w = \psi_s + \psi_p$ (2) $\psi_w = \psi_s$
(3) $\psi_w = \psi_p$ (4) $\psi_w = \psi_s - \psi_p$

115. Archaeobacteria are differ from other bacteria because:-

- (1) They have cellulosic cell wall
(2) They have nucleus
(3) They have different cell wall structure from other bacteria
(4) They are not found in salty areas

116. Which of the following is the most abundant protein in the whole of the biosphere ?

- (1) Collagen protein (2) Actin protein
(3) Albumin protein (4) Rubisco protein

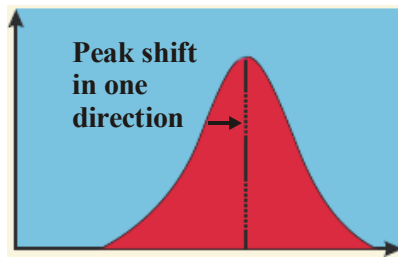
117. More distinct and clear annual rings are formed in temperate region plants, because ?

- (1) Climatic variation and temperature variation are not sharp.
(2) Temperature variations are sharp.
(3) Climate remains the same throughout the year.
(4) Climatic variations are sharp.

118. Which of the following formula can be used to find a value of vital capacity of lungs :-

- (1) IRV + ERV + TV
(2) IC + ERV
(3) EC + IRV
(4) All of the above are correct

119. Identify the type of natural selection shown below as well as related right environment in which this type of natural selection occurs :-



	Type of natural selection	Type of environment
(1)	Balancing	Constant environment
(2)	Directional	Continuous changing environment
(3)	Disruptive	Constant environment
(4)	Stablizing	Constant environment

120. Purpose of Immunisation is :-

- (1) Formation of Immunological memory
- (2) Strengthening innate Immunity
- (3) Promotion of phagocytosis
- (4) Activation of complementary system

121. Match the items of column I with column II and select the correct option :-

Column I		Column II	
(A)	Electrostatic precipitator	(1)	Removes gases like SO_2
(B)	Scrubber	(2)	Reduces automobile emission
(C)	Catalytic converter	(3)	Removes particulate matter

- (1) A – 2, B – 3, C – 1
- (2) A – 3, B – 2, C – 1
- (3) A – 1, B – 2, C – 3
- (4) A – 3, B – 1, C – 2

122. Which of the following gene can affect its inheritance in condition of a reciprocal cross :-

- (1) Plastid gene in *Mirabilis jalapa*
- (2) ABO blood group gene
- (3) Gene for shape of seed in pea plant
- (4) Lethal gene in mice

123. A man with pure black hair & blue eyes marries a woman with red hairs & pure black eyes. (Black hairs are dominant over red hairs and black eyes over blue eyes). What will be the probabilities of phenotypes of the children ?

- (1) 50% with black hairs & black eyes and 50% with black hairs & blue eyes
- (2) All will be with black eyes and black hairs
- (3) Both the above cases may be possible
- (4) 1 : 1 : 1 : 1

124. Guttation is most commonly observed under condition of ?

- (1) High atmospheric humidity and plentiful soil water
- (2) Low atmospheric humidity and little soil water
- (3) High atmospheric humidity and little soil water
- (4) Low atmospheric humidity and plentiful soil water

125. Gemmae are :-

- (1) Colourless, unicellular and asexual buds.
- (2) Green, multicellular and sexual buds.
- (3) Green, multicellular and asexual buds.
- (4) Green, unicellular and asexual buds.

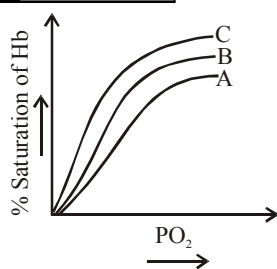
126. Which of the following is an example of saturated fatty acid ?

- (1) Palmitic acid
- (2) Oleic acid
- (3) Linoleic acid
- (4) Arachidonic acid

127. Bones have a hard and non pliable ground substance rich in (a) salts and (b) fibres :-

- (1) (a) Magnesium (b) Elastin
- (2) (a) Calcium (b) Collagen
- (3) (a) Phosphorous (b) Reticulin
- (4) (a) Calcium (b) Elastin

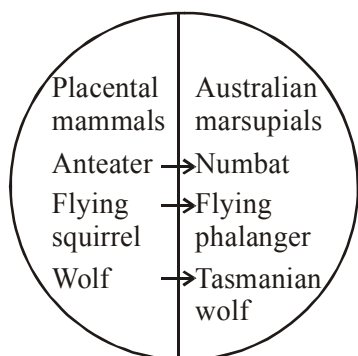
128.



Above shown graph represents oxyhaemoglobin dissociation curve. Find out the incorrect option related to graph :-

- (1) Shifting of B graph to C place is due to low CO_2 conc. and high pH value
- (2) If B graph represents adult Hb then C graph may be for fetal Hb
- (3) Affinity of Hb for O_2 is more for curve B as compared to curve A
- (4) P_{50} value will be more for graph C than graph A

129. Representd below is a type of evolution in organisms which type of evolution is and example of this pattern-



- (1) Convergent evolution
- (2) Divergent evolution
- (3) Retrogressive evolution
- (4) Quantum evolution

130. Which is not a secondary lymphoid organ :-

- (1) Thymus
- (2) Spleen
- (3) Appendix
- (4) MALT

131. Blood parasitism is very well explained by ?

- (1) Bacteriophage
- (2) Egg laying of cuckoo
- (3) Egg laying of crow
- (4) All of the above

132. Two genes R and Y are located very close on the chromosomal linkage map of maize plant. When RRYY and rryy genotypes are hybridized, the F_2 will show :-

- (1) Higher number of the parental types
- (2) Higher number of the recombinant types
- (3) Segregation in the expected 9 : 3 : 3 : 1 ratio
- (4) Segregation in 3 : 1 ratio

133. Find out the genotype of father and mother in the given pedigree chart :-

	Mother	Father
(1)	A A	A A
(2)	A a	A a
(3)	A A	a a
(4)	a a	A a

134. Find out the incorrect match of column I with column II :-

Column I	Column II
(1) Mg	Structural component of chlorophyll
(2) S	Found in some amino acid
(3) I	Important for plants
(4) Mn	Photolysis of water

135. Imperfect fungi is :-

- (1) Basidiomycetes
- (2) Deuteromycetes
- (3) Ascomycetes
- (4) Phycomycetes

136. Select the incorrect statement :-

- (1) In a normal flower, which opens and expose the anthers and stigmas complete autogamy is rather rare.
- (2) An ovule generally has a single embryosac.
- (3) Hilum is the junction of funicle and body of ovule.
- (4) Polar nuclei are situated above the egg apparatus in a ovule.

137. Match the list I with list II and find the correct option :-

	List-I [Epithelial tissue]		List II [Location]
(A)	Cuboidal	(i)	Epidermis of skin
(B)	Ciliated	(ii)	Inner lining of blood vessels
(C)	Columnar	(iii)	Inner surface of gall bladder
(D)	Squamous	(iv)	Inner lining of fallopian tube
(E)	Keratinized squamous	(v)	Lining of pancreatic duct

(1) A-v, B-iv, C-ii, D-iii, E-i

(2) A-iii, B-iv, C-v, D-ii, E-i

(3) A-v, B-iv, C-iii, D-ii, E-i

(4) A-iii, B-iv, C-v, D-i, E-ii

138. Read the following statements and count the numbers of wrong statements :-

(1) During joint diastole semilunar valves are closed.

(2) During artial systole the blood flow into the ventricle increase by about 30%.

(3) During joint systole AV valves are closed.

(4) Atrial phases are 0.3 sec later than the ventricular phases in a cardiac cycle.

(5) The cardiac output can be defined as the volume of blood pumped out by heart in each beat and average 5 liter in a healthy individual.

(1) 1 (2) 2 (3) 3 (4) 4

139. The inner ear contains a complex system located above the cochlea is A it is influenced by gravity and movements and helps us in B :-

(1) Organ of corti, Hearing

(2) Organ of corti, Maintaining balance

(3) Vestibular apparatus, Maintaining balance

(4) Eustachian tube, Hearing

140. How many of these drugs are obtained from plant:-

Cocaine, Morphine, Amphetamine, Cannabinoids, LSD, Datura.

Options :-

(1) Three (2) Four (3) Five (4) Six

141.  → Birds → Parasites

In above food chain energy transfer from tree to birds and from birds to parasites than what will be the shape of energy pyramid in this food chain ?

(1) Inverted pyramid

(2) Upright pyramid

(3) Spindle shape pyramid

(4) None of the above

142. In lac operon, which of the following gene function as constitutive gene and its product is available continuously ?

(1) Lac z gene

(2) Lac i gene

(3) Promotor gene

(4) Operator gene

143. Somatic hybridization is :-

(1) Fusion of male and female nucleus in a zygote

(2) Vegetation reproduction

(3) Fusion of two protoplast

(4) Sexual reproduction

144. (i) Promotes flowering in pineapple.

(ii) Used to prepare weed free lawn.

(iii) Promote the abscission of older leaves

The above function are carried out by :-

(1) GA

(2) C_2H_4

(3) ABA

(4) Auxin

145. Which is known as peat moss ?

(1) *Sphagnum*

(2) *Selaginella*

(3) *Marchantia*

(4) *Salvinia*

146. A safe place for laying eggs act as floral rewards in some plant/plants of angiosperms like :-

- (1) *Yucca* (2) *Amorphophallus*
(3) *Ficus* (4) All of them

147. In female cockroach, ootheca is secreted by :-

- (1) Oothecal chamber
(2) Oviduct
(3) Collateral gland
(4) Mushroom gland

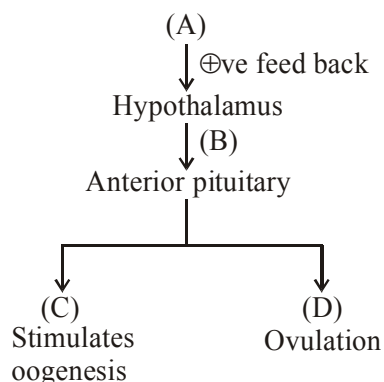
148. Find out the **incorrect** statement about E.C.G. :-

- (1) P wave represents the electrical excitation of the atria
(2) QRS complex represents the depolarization of the ventricle
(3) End of T wave represents the end of joint diastole
(4) By counting the number of QRS complexes that occur in a given time period any body can count the rate of heart beat

149. Which of the following hormone in body does not affect erythropoiesis :-

- (1) Cortisol (2) Thyroxine
(3) Erythropoietin (4) ADH

150.



Identify (A), (B), (C) and (D) hormones and select correct option :-

	A	B	C	D
(1)	Progesterone	GnRh	LH	FSH
(2)	Estrogen	GnRh	LH	FSH
(3)	Progesterone	LH	FSH	GnRh
(4)	Estrogen	GnRh	FSH	LH

151. Which one of the following ecosystem has the highest primary productivity ?

- (1) Pond ecosystem
(2) Lake ecosystem
(3) Grassland ecosystem
(4) Forest ecosystem

152. At a particular locus, frequency of 'A' allele is 0.6 and that of 'a' is 0.4. What would be the frequency of heterozygotes in a random mating population at equilibrium ?

- (1) 0.24 (2) 0.16
(3) 0.48 (4) 0.36

153. During anther culture, mature anthers are utilized for plant production. These anthers can also be referred as :-

- (1) Haploid plants (2) Explant
(3) Somatic embryo (4) Cybrid

154. Read the following statements :-

- (i) Each chromatid consists of DNA (long thread like) & histone
(ii) All secondary constrictions are called as NOR
(iii) In satellite region, only thymine base is absent
(iv) Telomere avoid joining of one chromosome to other

Identify incorrect statements :-

- (1) i, ii, iii (2) i, ii
(3) ii, iii (4) ii, iii, iv

155. Which of the following has more similarities in characters than family ?

- (1) Genus (2) Order
(3) Class (4) Division

156. Which of following structure play role in guiding the pollen-tube in to embryosac ?

- (1) Obturators
(2) Filiform apparatus
(3) Polar nuclei
(4) Antipodal cells

157. The best breeding method for animals that are below average in productivity in milk production and growth rate in beef cattle :-

- (1) Inbreeding
- (2) Interspecific hybridization
- (3) Out crossing
- (4) Cross breeding

158. Match the columns :-

	Column A		Column B
i	Accumulation of urea in blood	a	Glomerulonephritis
ii	Kidney transplanation	b	Renal stone
iii	Oxalates in kidney	c	Uraemia
iv	Inflammation of glomerulus	d	Renal failure

- (1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-b, iii-a, iv-d
(3) i-c, ii-d, iii-a, iv-b (4) i-c, ii-d, iii-b, iv-a

159. The class of adrenal gland hormones which are related with alertness, piloerection and more glucose level in blood is :-

- (1) Glucocorticoids (2) Catecholamines
- (3) Mineralocorticoids (4) Gonadocorticoids

160. What is the similarity between secondary oocyte and polar bodies:-

- (1) Volume of cytoplasm
- (2) Size of cell
- (3) Number of chromosomes
- (4) Both (1) and (3)

161. Match the followihng column-I with column-II ?

	Column - I		Column - II
(A)	Pyramid of number represents	(i)	Standing crop
(B)	Pyramid of biomass represents	(ii)	Productivity
(C)	Pyramid of energy represents	(iii)	Biotic potential

(1) A – (ii), B – (iii), C – (i)

(2) A – (iii), B – (i), C – (ii)

(3) A – (iii), B – (ii), C – (i)

(4) A – (ii), B – (i), C – (iii)

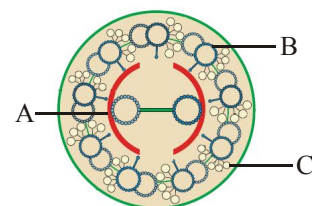
162. During 'Blue-White Colony' selection, blue colored colonies present in chromogenic substrate containing medium indicate :-

- (1) Transformed but non-recombinant
- (2) Transformed but recombinant
- (3) Non-transformed but recombinant
- (4) Non-transformed but non-recombinant

163. Which of the following option are correct regarding colour of pigments in chromatogram ?

	Chl-a	Chl-b	Xanthophyll	Carotenoids
(1)	Yellow orange	Yellow	Yellow green	Blue green
(2)	Blue green	Yellow green	Yellow orange	Only yellow
(3)	Yellow green	Yellow	Yellow orange	Blue green
(4)	Blue green	Yellow green	Yellow	Yellow to yellow orange

164. Identify A, B, C in following fig. :-



Options :-

	A	B	C
(1)	Central bridge	Dyenin	Actin
(2)	Central sheath	B Microtubule	Tubulin
(3)	Central sheath	A Microtubule	Dyenin
(4)	Central bridge	A Microtubule	Dyenin

165. Which of the following statements are true about *Laminaria* (kelps) ?

- (A) It is a rich source of iodine.
(B) Body divided into hold fast, stipe and lamina.
(C) Has pear shaped male gametes.
(D) Exhibits haplontic life cycle.
(E) Stored food is floridean starch.

- (1) A, B, C (2) A, C, D, E
(3) B, C, E (4) B, C, D

166. Underground stem store food in all the following except ?

- (1) Zaminkand (2) Colocasia
(3) Ginger (4) Sugarcane

167. Correctly match the column A with column B :-

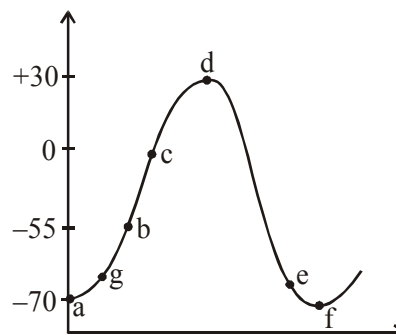
	Column A		Column B
1	Radula	a	Bony fishes
2	Air bladder	b	Coelenterates
3	Corals	c	Ctenophora
4	Comb plates	d	Mollusca

- (1) 1-a, 2-b, 3-c, 4-d (2) 1-d, 2-b, 3-a, 4-c
(3) 1-d, 2-a, 3-c, 4-b (4) 1-d, 2-a, 3-b, 4-c

168. Find out the incorrect match :-

(1)	Kidneys	Urea	25-30 gm/day
(2)	Skin	Sweat	NaCl, urea and lactic acid
(3)	Lungs	CO ₂	350 mL/min
(4)	Liver	Bile	Emulsification of fat

169. In given graph different points are labelled according to voltage and time. Identify them and give answers of following statements :-



- (i) At which point all -ve are cancelled by +ve during Na⁺ influx.
(ii) Where K⁺VGC opening is not possible.
(iii) Na⁺-K⁺ pump is closed between.
(iv) At which point K⁺VGC starts opening.

- | | | | |
|-------|------|-------|------|
| (i) | (ii) | (iii) | (iv) |
| (1) b | c | d-e | f |
| (2) c | b | g-d | d |
| (3) e | c | b-d | d |
| (4) c | b | g-d | b |

170. Which one of the following is an incorrect match:-

- (1) Lactational amenorrhoea - Natural contraception method
(2) Saheli - Steroidal pills
(3) Lippes loop - Non medicated IUCD
(4) Test tube baby - In vitro fertilization

171. According to Robert May global species diversity is about :-

- (1) more than 1.5 million
(2) 7 billion
(3) 7 million
(4) 7.5 million

172. Manipulation of DNA in genetic engineering became possible due to the discovery of :-

- (1) Restriction endonuclease
(2) DNA ligase
(3) Transcriptase
(4) Primase

173. How many of the following are associated with light reaction ?

- (i) Light Absorption
- (ii) ATP and NADPH+H formation
- (iii) Water splitting
- (iv) Regeneration of CO_2 acceptor
- (v) Oxidation of NADPH+H
- (vi) Oxygen release

- (1) Four (2) One
- (3) Two (4) Three

174. Arrange the following cell organelles on the basis of their size :-

- (A) Chloroplast (B) Ribosome
- (C) Lysosome (D) Nucleus

Options :-

- (1) $D > A > C > B$ (2) $C > D > B > A$
- (3) $A > D > C > B$ (4) $D > A > B > C$

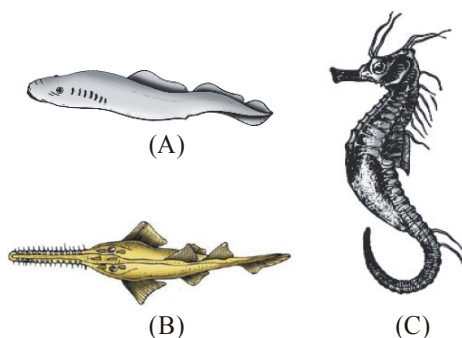
175. Green algae usually have a rigid cellwall made up of an outer layer of :-

- (1) Cellulose (2) Pectose
- (3) Lignin (4) Hemicellulose

176. Mustard and groundnut are :-

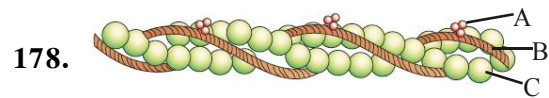
- (1) Dry fruits
- (2) Fleshy fruits
- (3) Dry and dehiscent fruits
- (4) Fleshy and dehiscent fruits

177. Figures of some animals are given below -



Which one of the following is **incorrect** for the above figure ?

- (1) A-has 6-15 pairs gill slits
- (2) B-has air bladder
- (3) C-has 4 pairs of gills
- (4) B-has placoid scales



Identify the correct labelling :-

- (1) A-Tropomyosin, B-Troponin, C-F-actin
- (2) A-F-actin, B-Troponin, C-Tropomyosin
- (3) A-Troponin, B-Tropomyosin, C-F-actin
- (4) A-F-actin, B-Tropomyosin, C-Troponin

179. Maximum digestion of food occurs in A and maximum absorption occurs in B the part of intestine

- (1) Ileum, Jejunum
- (2) Jejunum, Duodenum
- (3) Duodenum, Jejunum
- (4) Ileum, Duodenum

180. Match column I and column II :

	Column I		Column II
(A)	Bile juice	(i)	Nucleases
(B)	Saliva	(ii)	Aminopeptidases
(C)	Pancreatic juice	(iii)	Ptyalin
(D)	Succus entericus	(iv)	Na-glycholate

and choose correct option :-

- (1) A-iv, B-iii, C-i, D-ii
- (2) A-iii, B-iv, C-i, D-ii
- (3) A-iv, B-ii, C-i, D-iii
- (4) A-iv, B-iii, C-ii, D-i